Entity-aware financial headlines sentiment analysis

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Abstract

Financial news headlines and social media content play an important role in shaping market expectations and influencing asset prices. This makes financial sentiment analysis a key task, particularly when the objective is to understand sentiment toward specific entities.

In this work, we study entity-aware sentiment analysis in financial news headlines. Unlike standard approaches that assign a single sentiment to an entire text, this setting requires identifying the sentiment expressed toward a given target entity, which becomes more challenging when multiple entities with potentially conflicting sentiments are present.

We build on the SEntFiN 1.0 framework, which provides an entity-level annotated dataset of financial headlines. We compare a range of natural language processing approaches, from lexical representations such as TF-IDF to latent semantic representations and transformer-based models. In addition, we evaluate how these models generalize from single-entity to multi-entity settings.

Our results show that while classical models perform adequately in simple settings, their performance degrades significantly in multi-entity scenarios. In contrast, transformer-based models remain more robust, highlighting the importance of contextual representations for capturing fine-grained financial sentiment.

1 Introduction

Financial news and headlines play a central role in modern financial markets, as they provide timely information that can influence investors' expectations and, ultimately, asset prices.

Financial sentiment analysis aims to determine the polarity of information conveyed in textual data, typically classifying it as positive, negative, or neutral. In practice, however, this task is more complex than standard sentiment analysis. Financial headlines often mention multiple entities—such as companies, sectors, or institutions—and may express different, sometimes conflicting, sentiments toward each of them. As a result, assigning a single sentiment to an entire headline can lead to a loss of information and misinterpretation.

To address this issue, recent work has focused on entity-aware sentiment analysis, where the objective is to extract the sentiment associated with a specific target entity within a given text. In this context, the SEntFiN 1.0 framework introduces a dataset of financial news headlines from the Indian market, annotated at the entity level, and proposes a methodology to handle headlines containing multiple entities with potentially conflicting sentiments.

This project is situated within this framework and focuses on the task of entity-aware sentiment classification for financial news headlines. In particular, we consider both single-entity and multi-entity headlines and investigate how different natural language processing techniques perform in this setting. We compare several text representations, ranging from lexical approaches such as TF-IDF to latent semantic representations and contextual embeddings obtained from transformer-based models.

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

The goal is to understand how the choice of representation impacts the ability to capture nuanced financial sentiment, especially in the presence of multiple entities.

2 Litterature review

Financial sentiment analysis (FSA) has become an important application of natural language processing in finance, as textual information from news, reports, and social media plays a key role in shaping market expectations and price dynamics. Early approaches relied primarily on lexicon-based methods, such as sentiment dictionaries, which assign polarity scores to words and aggregate them at the sentence or document level. However, these approaches often fail to capture the contextual and domain-specific meaning of financial language, leading to inaccurate sentiment estimation.

With the development of machine learning, researchers introduced supervised models trained on annotated datasets. These methods improved performance but were still limited by the availability of labeled data and their inability to handle complex linguistic structures. More recently, deep learning and transformer-based models, such as FinBERT, have significantly improved sentiment classification by leveraging contextual embeddings and domain-specific pretraining.

A key limitation of most existing work is that sentiment is typically assigned at the sentence level, assuming a single sentiment per text. This assumption is unrealistic in financial contexts, where headlines frequently mention multiple entities with potentially conflicting sentiments. To address this issue, recent research has moved toward fine-grained or entity-level sentiment analysis, where the goal is to extract sentiment with respect to a specific entity.

Within this line of work, the SEntFiN 1.0 (1) framework constitutes a significant contribution. It introduces a large human-annotated dataset of over 10,000 financial news headlines with entity-level sentiment labels, including many cases with multiple entities and conflicting sentiments. The paper proposes a simple yet effective transformation—replacing entities with Target and Other tokens—to enable models to focus on entity-specific sentiment without relying on complex linguistic parsing. In addition, it provides a comprehensive comparison of modeling approaches, ranging from lexicon-based methods to pretrained language models. The results show that transformer-based models such as RoBERTa and FinBERT achieve the best performance, with accuracy above 94

Overall, this work bridges the gap between traditional sentence-level sentiment analysis and more realistic financial scenarios involving multiple entities.

3 Dataset analysis

The SEntFiN 1.0 dataset is composed of 10,753 annotated news headlines, of which 2,847 refer to multiple entities. Among these, 1,233 contain conflicting sentiments. This results in a total of 14,404 entity–sentiment pairs. When multiple entities are present, there are only two entities in 78.67% of the cases, although a sentence can contain up to seven entities.

The average length of a news headline is 9.91 words. Sentiments are categorized into three classes: positive, negative, and neutral. The dataset does not suffer from strong class imbalance, with the positive class representing 35.22% of the data and the negative class 26.47%.

Following the SEntFiN 1.0 framework, each multi-entity headline is duplicated such that, in each instance, one entity is replaced by `TARGET` and the other entities by `OTHER`. The same transformation is applied to single-entity headlines. We observe that the `TARGET` token can appear in any position in the sentence, although it is more frequently located at the beginning.

In order to perform further analysis, we construct a vector representation of the text. We start with a TF-IDF representation, which transforms the collection of documents into a matrix of features. Each feature corresponds to a word, weighted by the product of its term frequency (capturing local importance within a document) and its inverse document frequency (capturing global rarity across the corpus).

We construct a custom stopword set, removing common stopwords while keeping the token `OTHER`. The vocabulary is limited to 10,000 features. The score assigned to each word is computed as the average value of the corresponding TF-IDF feature across all documents.

We observe that `TARGET` and `OTHER` have the highest scores, followed by financial terms such as *rs*, *crore*, and *net*. This reflects their high frequency across the dataset.

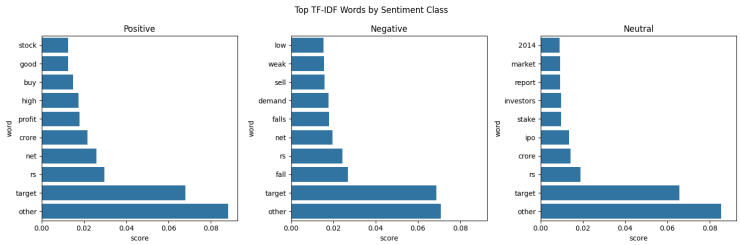

Figure 1: Top TF-IDF words by sentiment class

We then analyze the most important features per sentiment class. Positive and negative sentiments are mainly driven by directional tokens, such as *profit* or *high* for positive sentiment, and *fall* or *low* for negative sentiment. In contrast, neutral samples are characterized by more general financial terminology, such as *IPO*, *investors*, or *market*. Some tokens, such as *net*, *rs*, or *crore*, appear across all sentiment classes, indicating that they are not discriminative.

We then perform a Latent Semantic Analysis using a truncated Singular Value Decomposition (SVD) applied to the TF-IDF matrix. The resulting latent space exhibits several distinct branches, each predominantly associated with a sentiment class. However, the separation is not fully clear, as the branches overlap significantly.

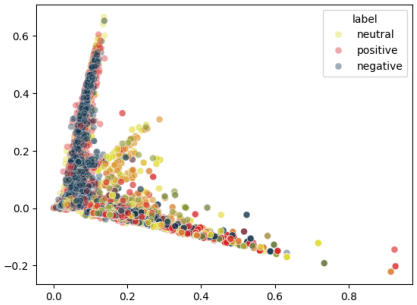

Figure 2: PCA projection of TF-IDF features

We interpret the 50 dimensions obtained from SVD as latent topics. To better understand their relationship with sentiment, we compute the mean value of each topic per class. After standardization, we observe that several topics align with sentiment polarity, for instance topics 14 and 15, which show clear differences across classes and a progression from negative to positive.

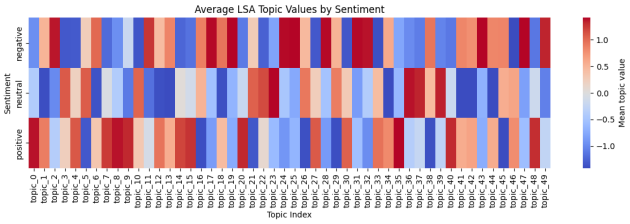

Figure 3: Average LSA topic values by sentiment

4 Experiments

4.1 Modeling and Experimental Setup

We design two main experiments to evaluate the impact of entity-aware annotation across different text representations and classification models.

First, we perform a standard random train–test split, where both single-entity and multi-entity samples are mixed. This setting allows us to assess the overall predictive performance of the models under a classical supervised learning framework.

Second, we explicitly test the robustness of the models to multi-entity settings by training only on single-entity samples and evaluating on multi-entity samples. This experiment is designed to measure how well models generalize to more complex sentences containing multiple entities with potentially conflicting sentiments.

We consider several text representations of increasing complexity. We start with TF-IDF features, which provide a sparse lexical representation of the text. To capture higher-level semantic structure, we also project these features into a lower-dimensional latent space using truncated Singular Value Decomposition (LSA).

On top of these representations, we train classical machine learning models, namely Logistic Regression and Gradient Boosting.

Finally, we use a transformer-based model, FinBERT, which produces contextualized embeddings of the text. Unlike TF-IDF or LSA representations, FinBERT captures the meaning of words depending on their context within the sentence. We fine-tune this model directly on our dataset for the entity-level sentiment classification task. This approach allows us to evaluate whether contextual representations can better handle the complexity of financial language, especially in the presence of multiple entities.

4.2 Results

We report the performance of the different models under two experimental settings: a standard mixed train–test split and a transfer setting where models are trained on single-entity samples and evaluated on multi-entity samples.

Model	Accuracy	F1 (Macro)	F1 (Weighted)
finBERT	0.881	0.882	0.881
TF-IDF + LR	0.793	0.793	0.793
LSA + LR	0.681	0.678	0.680
LSA + XGB	0.694	0.688	0.693

Table 1: Performance on mixed train–test split

In the mixed setting, FinBERT significantly outperforms all other models, reaching an accuracy of 88%. Classical models based on TF-IDF also perform relatively well, indicating that lexical information already captures a large part of the sentiment signal. In contrast, LSA-based models underperform, suggesting that dimensionality reduction leads to a loss of discriminative information.

Model	Accuracy	F1 (Macro)	F1 (Weighted)
FinBERT	0.851	0.853	0.852
TF-IDF + LR	0.616	0.602	0.609
LSA + LR	0.567	0.558	0.557
LSA + XGB	0.529	0.504	0.504

Table 2: Train on single-entity, test on multi-entity samples

Training on single-entity samples and evaluating on multi-entity samples leads to a significant drop in performance for all models. This highlights the increased difficulty of multi-entity sentiment extraction, where models must correctly attribute sentiment to a specific target in the presence of multiple entities and potentially conflicting signals.

The performance degradation is particularly strong for classical models. TF-IDF combined with Logistic Regression drops from around 79% accuracy in the mixed setting to 61%, while LSA-based models fall below 60%. This indicates that lexical and low-dimensional representations struggle to generalize to more complex sentence structures, even with Gradient Boosting.

In contrast, FinBERT remains significantly more robust, with only a moderate decrease in performance. This suggests that contextual representations are better suited to capture interactions between entities and disentangle conflicting sentiment signals.

Overall, these results confirm that while classical lexical models can perform well in simple settings, they struggle to generalize to more complex structures. Transformer-based models, by leveraging contextual information, provide a clear advantage in handling multi-entity financial text.

4.3 Error analysis

Standard mixed training Inspecting the confusion matrices shows that TF-IDF + LR errors are mainly due to confusion of true positive or negative samples with the neutral class. The lack of contextual information is evident in headlines such as *TARGET narrows down net loss to Rs 156.36 crores*. The words *narrows* and *down* lead the model to predict a negative sentiment with high confidence, whereas the true label is positive.

Similar conclusions can be drawn for LSA + LR. However, it shows a stronger degradation: it confuses 128 true negatives as positive and 142 as neutral. In contrast, TF-IDF confuses 111 true negatives as neutral and only 53 as positive. This suggests that the latent representation reduces the model's ability to clearly separate negative and positive classes, while still struggling with neutral.

LSA + XGB does not show significant improvements, highlighting that the poor performance of LSA + LR is not due to the simplicity of the classifier but rather to the latent representation itself.

FinBERT, however, achieves the best overall performance and very rarely confuses negative with positive (22 cases) or positive with negative (37 cases). Its errors mostly occur in genuinely ambiguous headlines, such as *TARGET posts Q3 net loss of Rs 22.25 cr; stock up*. The model predicts a positive sentiment due to *stock up*, whereas the true label is negative. This suggests that, despite strong performance, FinBERT still struggles with sentences containing conflicting signals within the same headline.

More generally, FinBERT tends to focus on the most salient expression in the sentence, without fully resolving which financial concept should dominate the sentiment.

Cases where TF-IDF is correct while FinBERT is wrong typically correspond to sentences where sentiment depends on financial context. For example, *TARGET biggies earnings belies expectations* is labeled as positive because expectations were low. TF-IDF predicts correctly, likely due to the presence of the word *expectations*, while FinBERT predicts a negative sentiment.

Transfer training: single to multiple The performance of TF-IDF degrades significantly in this setting. In particular, the model predicts almost as many neutral labels as positive for true positive samples, and similarly for negative samples. This indicates a loss of discriminative power.

A very similar behavior is observed for LSA + LR and LSA + XGB. In both cases, the models more frequently misclassify positive and negative samples as neutral than correctly identifying them. This suggests that, when faced with multiple entities and potentially conflicting signals, the models tend to default to the neutral class.

This behavior can be explained by the fact that these models were trained only on single-entity samples and therefore never learned to handle the `OTHER` token or disentangle multiple sentiment signals.

FinBERT remains significantly more robust in this transfer setting. By leveraging contextual representations, it is better able to capture interactions between entities and identify the relevant sentiment, even in more complex sentences.

Figure 4: Error rate by sentiment class across models. The neutral class appears to be the most difficult to classify, especially for classical models, while finBERT achieves consistently lower error rates across all classes.

5 Conclusion and Discussion

Overall, the results highlight that the main difficulty of the task does not lie in sentiment classification itself, but in correctly attributing sentiment in the presence of multiple entities and potentially conflicting signals. While all models achieve reasonable performance in the standard mixed setting, their behavior diverges significantly when evaluated under transfer settings.

Classical approaches based on TF-IDF and LSA show strong limitations. Their performance relies heavily on lexical cues and they fail to capture the structure of the sentence. As a result, they struggle both with contextual polarity (e.g., improvements despite negative words such as *loss*) and with multi-entity settings. In the transfer experiment from single to multiple entities, these models tend to default to the neutral class, suggesting that they are unable to disentangle multiple sentiment signals when they have not been explicitly exposed to such patterns during training. This behavior indicates a lack of robustness to distribution shifts in sentence structure.

LSA-based representations further degrade performance. While they aim to capture latent semantic structure, they appear to blur polarity information, reducing the separability between positive and negative classes. The fact that XGBoost does not significantly improve over Logistic Regression confirms that the limitation comes from the representation rather than the classifier.

In contrast, FinBERT consistently achieves strong performance across all settings and remains significantly more robust under transfer. Its ability to leverage contextual representations allows it to better handle interactions between entities. However, the error analysis shows that it is not immune to failure. In particular, it struggles with genuinely ambiguous headlines and with sentences containing conflicting signals, where different parts of the sentence suggest opposite sentiments.

Another limitation of our approach lies in the data transformation itself. Replacing entities with `TARGET` and `OTHER` simplifies the task and makes it more tractable, but it also removes potentially useful information such as the identity of the entity or sector-specific context. This may artificially reduce the difficulty of the problem and does not fully reflect real-world applications where entity identity matters.

Furthermore, the evaluation remains limited to a single dataset and a specific annotation scheme. Although the transfer experiments provide some insight into robustness, they do not fully capture real-world distribution shifts, such as changes in writing style, market regimes, or domain. The relatively small size of the dataset, especially for highly multi-entity sentences, also limits the statistical strength of some conclusions.

Finally, while FinBERT shows strong empirical performance, it remains a black-box model. Unlike TF-IDF + LR, which allows for direct interpretation of features, the decision process of FinBERT is less transparent. This raises questions regarding interpretability and reliability in financial applications, where understanding model decisions can be important.

Overall, the results suggest that contextual models are necessary to handle multi-entity financial sentiment analysis, but they still face challenges in resolving conflicting information and incorporating deeper financial reasoning. Improving robustness would likely require both richer data and models that explicitly account for entity interactions and financial context.

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